

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10: 4.1 STATISTICS I

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Madam Achieng, the Deputy Principal of a day secondary school in Kisumu, has given the Grade 10 class the test scores of the same 20 learners in Subject 1 (Mr. Otieno) and Subject 2 (Mrs. Wanjiku). The pass mark is 50 out of 100. Answer every question in the space under it, using the data from our lessons. Show your working for every calculation. Write grouped-data answers as estimates and give reasons in full sentences. When you talk about WHY the scores differ, say what we cannot yet know and use words such as 'may', 'could' and 'we need more data'. Your answers should help you answer the driving question: How can you use the two sets of data to predict the learners' and the teachers' next steps?

Part 1: Organising the scores into frequency tables

Here are the scores in ascending order.

Subject 1: 67, 69, 70, 72, 73, 74, 76, 77, 78, 80, 81, 82, 84, 85, 86, 88, 89, 90, 91, 95.

Subject 2: 11, 13, 15, 18, 21, 23, 27, 29, 30, 31, 32, 32, 35, 42, 44, 45, 51, 58, 60, 60.

(a) Using a common class width of 10, with classes written 10-19, 20-29, ... up to 90-99, make one grouped frequency table for both subjects. Use tally marks grouped in fives and show the frequency for each subject. Check your totals.

(b) Why did we use the same classes for both subjects?

(c) If you made an ungrouped frequency table for Subject 1,

(a) Frequencies (the tally marks match these numbers, written in groups of five):

Class	Subject 1 f	Subject 2 f
10-19	0	4
20-29	0	4
30-39	0	5
40-49	0	3
50-59	0	2
60-69	2	2
70-79	7	0
80-89	8	0
90-99	3	0
Total	20	20

what would you notice, and why does grouping help Madam Achieng more? (Key Inquiry 1: How do we organise data using frequency distribution tables?)	Both frequency columns add up to $n = 20$, so every score has been counted once. (b) Using the same classes lets us compare the two subjects fairly, because each row stands for the same marks in both columns. (c) In an ungrouped table for Subject 1, every score occurs once (20 different scores, each $f = 1$). The table would be as long as the list and would show little. Grouping gives 9 short rows that show where the scores cluster: 15 of the 20 Subject 1 scores are in 70-89, while Subject 2 scores are spread from 10-19 to 60-69, with 16 of 20 below 50. A busy teacher can see this pattern at a glance.
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Part 2: Finding the typical score	
Use the ordered lists in Part 1 (the sum of the Subject 1 scores is 1607 and the sum of the Subject 2 scores is 677). (a) Calculate the mean and the median of each subject. (b) State the mode of each subject, and explain what it means when a set has no mode or two modes. (c) Compare the typical score in each subject with the pass mark of 50. What does this suggest about whether the learners are weak overall?	(a) Subject 1: mean = $1607 \div 20 = 80.35$. For the median, $n = 20$, so we average the 10th and 11th ordered values: $(80 + 81) \div 2 = 80.5$. Subject 2: mean = $677 \div 20 = 33.85$. Median = $(31 + 32) \div 2 = 31.5$. (b) Subject 1 has no mode, because every score occurs only once. Subject 2 has two modes, 32 and 60, because each occurs twice and every other score occurs once; we say it is bimodal. A mode is the most frequent score, so a set can have none, one or more than one. (c) The typical score in Subject 1 (about 80) is well above the pass mark of 50. The typical score in Subject 2 (about 32 to 34) is below 50. The same 20 learners scored high in one subject and low in the other, so they cannot all be weak. The difference seems to be between the subjects. These measures do not tell us why the scores differ, and they do not show how spread out the scores are.

Part 3: Reading position and spread from cumulative frequency										
The grouped frequencies are:			(a) Subject 2:							
Class	Subject 1 f	Subject 2 f	Score less than	9.5	19.5	29.5	39.5	49.5	59.5	69.5
10-19	0	4	cf	0	4	8	13	16	18	20
20-29	0	4								
30-39	0	5								
40-49	0	3								
			(b) Q1 is the 5th learner. It lies in 20-29 (cf before = 4, f = 4): $Q1 = 19.5 + (1/4) \times 10 = 22.0$. The median is the 10th learner. It lies in 30-39 (cf before = 8, f = 5): median = $29.5 + (2/5) \times 10 = 33.5$.							

50-59	0	2
60-69	2	2
70-79	7	0
80-89	8	0
90-99	3	0

(a) Make a cumulative frequency table for Subject 2 using the upper class boundaries (less-than values).

(b) Using the 5th, 10th and 15th learners, estimate Q1, the median and Q3 for Subject 2, and its interquartile range (IQR). Show your working.

(c) For Subject 1 the estimates are Q1 = 73.8 and Q3 = 87.0. Find its estimated IQR and compare it with that of Subject 2. What does this tell you about the spread of the middle half of the learners in each subject?

(d) Why are these values called estimates?

Q3 is the 15th learner. It lies in 40-49 (cf before = 13, f = 3): $Q3 = 39.5 + (2/3) \times 10 = 46.2$ (1 dp).
Estimated IQR = $46.2 - 22.0 = 24.2$.

(c) Subject 1 estimated IQR = $87.0 - 73.8 = 13.2$. Subject 2 has the larger IQR (24.2 compared with 13.2), so the middle half of the Subject 2 scores is more spread out, while the middle half of the Subject 1 scores is close together. This agrees with the ranges: 28 for Subject 1 and 49 for Subject 2.

(d) They are estimates because the grouped table does not show the exact scores. We only know how many scores fall in each class, so we assume the scores are spread evenly within the class. They differ slightly from the exact values from the ungrouped lists, for example 33.5 compared with the exact median of 31.5.

Part 4: Claim, evidence and reasoning

A classmate says: 'These 20 learners are all weak, so Madam Achieng should tell the whole class to repeat the term.'

(a) Write a claim-evidence-reasoning response. Use at least three pieces of evidence from the tables, measures or graphs (for example pass counts, a measure of centre and a measure of spread).

(a) Claim: The learners are not all weak, so the whole class should not repeat the term. The difference is between the subjects.
Evidence: All 20 learners passed Subject 1 (lowest score 67, mean 80.35, median 80.5), but only 4 of 20 passed Subject 2 (16 scored below 50, mean 33.85, median 31.5). The range is 28 in Subject 1 and 49 in Subject 2.

Reasoning: The same 20 learners did very well in one subject. If they were all weak, they would not all pass Subject 1 with scores of 67 or more. So the low Subject 2 results do not describe the learners themselves. Decisions need to be made for each subject separately.

(b) The histograms and polygons show the shape at a glance. Subject 1 is a tight block of bars between 60 and 99, tallest at 80-89 (f = 8). Subject 2 is spread across 10 to 69, tallest at 30-39 (f = 5), and sits much further left. The raw lists were just long rows of numbers. The graphs show the centre, the spread and the position of each subject. They still cannot tell us why the scores differ.

(b) We saw that Subject 1 histograms and polygons are narrow and to the right, and Subject 2 histograms and polygons are wide and to the left. Explain what this shape tells you that the raw lists did not.	
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Part 5: Recommendations and why data helps decisions	
Madam Achieng wants a final report that answers the driving question: How can you use the two sets of data to predict the learners' and the teachers' next steps? (a) Write one evidence-based recommendation each for Mr. Otieno, for Mrs. Wanjiku and the learners, and for the 16 learners who scored below 50 in Subject 2. Link each to the data. (b) Name two reasons that could explain the difference in scores. Explain why you cannot yet claim that either is the cause, and what further data you would collect. (c) Answer Key Inquiry 2: Why is it important to collect, organise and interpret data in making decisions? Use what changed between the raw lists and your tables, measures and graphs.	(a) Mr. Otieno: Continue with the Subject 1 teaching and give extension or enrichment work, because all 20 learners passed and the scores are high and close together (mean 80.35, range 28). Mrs. Wanjiku and the learners: Reteach and revise the Subject 2 topics, because only 4 of 20 passed, the mean is 33.85 and the scores are widely spread (range 49). The 16 learners below 50 in Subject 2: Run remedial support for them, as they failed at the pass mark of 50. Seven of the 20 scored 40 or more, so some are close to the pass mark. We predict that these steps could help, but we should also collect more data. (b) The test in Subject 2 may have been harder, or the teaching method or revision time may have been different. The data we have only shows what the scores were. It does not show the difficulty of the test, the way each subject was taught or the time spent revising, so these are hypotheses to investigate, not facts. We would collect more data, such as topic-by-topic marks and a second test, to find the cause before concluding. (c) The raw and ordered lists only showed the highest, lowest and pass or fail. They hid the patterns, the spread and what action was needed. When we organised the data into frequency tables and interpreted it with measures and graphs, we could see that Subject 1 is high and tightly clustered while Subject 2 is low and widely spread. This lets us make decisions for each subject, support our recommendations with evidence, and decide what extra data is still needed.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Organising data (frequency tables, Key Inquiry 1)	Builds a complete, correct grouped table with a common class width of 10, correct	Builds a mostly correct table with one small error, and gives a basic reason for	Table has several errors or totals that do not equal 20, or gives no reason for the

	tallies and frequencies totalling 20 for each subject, and clearly explains why grouping is more useful than an ungrouped table.	using the same classes or for grouping.	choice of classes.
Calculating measures (mean, median, mode, quartiles, IQR)	All measures are correct (80.35, 80.5, 33.85, 31.5, modes 32 and 60, IQR estimates 13.2 and 24.2) with clear working, and grouped values are called estimates with a reason given.	Most measures are correct with working shown, with a small slip, or the reason for 'estimate' is incomplete.	Several measures are incorrect or working is missing, and does not recognise grouped values as estimates.
Interpreting tables, measures and graphs	Uses centre, spread, shape and pass rate together to compare the subjects, and explains what the graphs show that the raw lists hid.	Uses at least two features (such as the mean and the pass count) to compare the subjects, with limited reference to spread or shape.	Only lists numbers or describes one subject, with no comparison.
Claim-evidence-reasoning and recommendations	Gives a clear claim supported by several correct pieces of evidence, and recommendations for Mr. Otieno, Mrs. Wanjiku, the learners and the 16 learners below 50, each linked to the data.	Gives a claim with some evidence and recommendations for most groups, but the link between the data and the recommendation is sometimes weak.	Claim is unsupported or recommendations are missing or not linked to the data.
Separating what the data shows from what it cannot explain	Treats reasons for the score difference as hypotheses using words such as 'may' and 'could', and names suitable extra data (topic-by-topic marks, a second test).	Recognises that causes are not known but gives an unclear or incomplete plan for finding out.	States a cause for the score difference as a fact.

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